p-values and Neyman-Pearson test

Ciaran Evans

Recap: p-values

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

A **p-value** $p(X_1, ..., X_n)$ is a test statistic satisfying:

- $0 \leq p(X_1, ..., X_n) \leq 1$
- Reject H_0 for small values of $p(X_1, ..., X_n)$ (e.g. reject when $p \leq \alpha$)
- For all $\theta \in \Theta_0$, $P_\theta(p(X_1, ..., X_n) \leq \alpha) \leq \alpha$

If I reject when $p \leq \alpha$, then

$P(\text{type I error} \mid H_0) \leq \alpha$

p-values

Suppose we have a test which rejects H_0 when $T(X_1, ..., X_n) > c_\alpha$, where c_α is chosen so that

$$\sup_{\theta \in \Theta_0} \beta(\theta) = \sup_{\theta \in \Theta_0} P_\theta(T(X_1, ..., X_n) > c_\alpha) \leq \alpha$$

Let $x_1, ..., x_n$ be a set of observed data.

biggest probability under H_0

Theorem: Let

$$p(x_1, ..., x_n) = \sup_{\theta \in \Theta_0} P_\theta(T(X_1, ..., X_n) > \underbrace{T(x_1, ..., x_n)}_{\text{observed}})$$

test statistic

Then p is a valid p-value.

"probability of observed test statistic or 'more extreme', under H_0"

WTS $P_\theta(p(X_1, \dots, X_n) \leq \alpha) \leq \alpha$ $\forall \alpha \in (0,1]$ and $\forall \theta \in \Theta_0$

Proof of theorem

$X = (X_1, \dots, X_n)$ (r.v.s) $\qquad x = (x_1, \dots, x_n)$ (obs. data)

Let $\theta \in \Theta_0$, $\alpha \in [0,1]$, and let $F_\theta(t)$ be cdf of $T(X)$. Proof in case that F_θ is continuous $\forall \theta$

$$p_\theta(x) = P_\theta(T(X) > T(x)) = 1 - P_\theta(T(X) \le T(x))$$
$$= 1 - F_\theta(T(x))$$

$$P_\theta(p_\theta(X) \le \alpha) = P_\theta(1 - F_\theta(T(X)) \le \alpha)$$
$$= P_\theta(F_\theta(T(X)) \ge 1 - \alpha)$$
$$= P_\theta(T(X) \ge F_\theta^{-1}(1-\alpha))$$
$$= 1 - F_\theta(F_\theta^{-1}(1-\alpha))$$
$$= 1 - (1-\alpha) = \alpha$$

$$\Rightarrow P_\theta(p_\theta(X) \le \alpha) \le \alpha \qquad \forall \theta$$

$$p(X) = \sup_{\theta' \in \Theta_0} p_{\theta'}(X) \ge p_\theta(X)$$

$$\Rightarrow P_\theta(p(X) \le \alpha) \le P_\theta(p_\theta(X) \le \alpha) \le \alpha \qquad //$$

Recap: hypothesis testing and power function

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

Given observed data $X_1, ..., X_n$:

1. Calculate a test statistic $T_n = T(X_1, ..., X_n)$
2. Choose a rejection region $\mathcal{R} = \{(x_1, ..., x_n) : \text{ reject } H_0\}$
3. Reject H_0 if $(X_1, ..., X_n) \in \mathcal{R}$ (or reject if $p \leq \alpha$)

The **power function** $\beta(\theta)$ is

$$\beta(\theta) = P_\theta((X_1, ..., X_n) \in \mathcal{R})$$

Goal: maximize power for $\theta \in \Theta_1$, subject to control of power for $\theta \in \Theta_0$

Wald test for normal mean

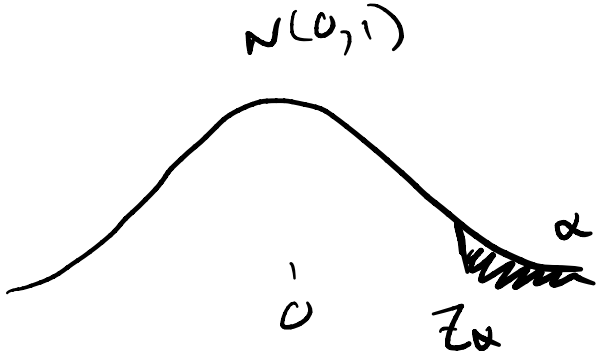
Let $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known. We wish to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu = \mu_1$$

where $\mu_1 > \mu_0$. The Wald test rejects if

look in right tail

$$\frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} > z_\alpha$$



$$\Rightarrow \bar{X}_n > \underbrace{\mu_0 + \frac{\sigma}{\sqrt{n}} z_\alpha}_{c_0}$$

(chosen such that $P(\text{type I error} \mid H_0) = \alpha$)

Another angle

Let $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known. We wish to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu = \mu_1$$

where $\mu_1 > \mu_0$.

Suppose we reject H_0 when

$$\frac{L(\mu_1|X_1, ..., X_n)}{L(\mu_0|X_1, ..., X_n)} = \frac{f(X_1, ..., X_n|\mu_1)}{f(X_1, ..., X_n|\mu_0)} > k_0$$

For $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$:

$$\frac{L(\mu_1 | X_1, \dots, X_n)}{L(\mu_0 | X_1, \dots, X_n)} = \frac{\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^n (X_i-\mu_1)^2\right\}}{\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^n (X_i-\mu_0)^2\right\}}$$

$$= \exp\left\{-\frac{1}{2\sigma^2}\left[\sum_{i=1}^n (X_i-\mu_1)^2 - \sum_{i=1}^n (X_i-\mu_0)^2\right]\right\}$$

$$\frac{L(\mu_1 | X_1, \dots, X_n)}{L(\mu_0 | X_1, \dots, X_n)} > k_0 \iff -\frac{1}{2\sigma^2}\left(\sum_{i=1}^n (X_i-\mu_1)^2 - \sum_{i=1}^n (X_i-\mu_0)^2\right) > \log k_0$$

Rearranging

$$\frac{L(\mu_1 | x_1, \dots, x_n)}{L(\mu_0 | x_1, \dots, x_n)} > k_0 \iff -\frac{1}{2\sigma^2}\left(\sum_{i=1}^n (x_i-\mu_1)^2 - \sum_{i=1}^n (x_i-\mu_0)^2\right) > \log k_0$$

$$\iff \underbrace{\sum_{i=1}^n (x_i-\mu_1)^2}_{\sum_{i=1}^n (x_i^2 - 2x_i\mu_1 + \mu_1^2)} - \sum_i (x_i-\mu_0)^2 < -2\sigma^2 \log k_0$$

$$\hookrightarrow \sum_{i=1}^n x_i^2 - 2n\bar{x}\mu_1 + n\mu_1^2$$

$$\iff \sum_i x_i^2 - 2n\bar{x}\mu_1 + n\mu_1^2 - \sum_i x_i^2 + 2n\bar{x}\mu_0 - n\mu_0^2 < -2\sigma^2 \log k_0$$

$$\iff 2n\bar{x}(\mu_0 - \mu_1) < -2\sigma^2 \log k_0 - n(\mu_1^2 - \mu_0^2)$$

$$\iff \bar{x} > \underbrace{\frac{-2\sigma^2 \log k_0 - n(\mu_1^2 - \mu_0^2)}{2n(\mu_0 - \mu_1)}}_{c_0}$$

So: For $x_i \overset{iid}{\sim} N(\mu, \sigma^2)$, $\dfrac{L(\mu_1 | x_1, \dots, x_n)}{L(\mu_0 | x_1, \dots, x_n)} > k_0$

$\iff \bar{x} > c_0$ (where we can choose c_0 to get desired type I error rate)

Rearranging

Let $\mathbf{X} = X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known. We wish to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu = \mu_1$$

where $\mu_1 > \mu_0$.

The Wald test rejects if $\overline{X}_n > c_0$, which is equivalent to rejecting when

$$\frac{L(\mu_1|\mathbf{X})}{L(\mu_0|\mathbf{X})} = \frac{f(X_1, ..., X_n|\mu_1)}{f(X_1, ..., X_n|\mu_0)} > k_0$$

Neyman-Pearson test

Let $\mathbf{X} = (X_1, ..., X_n)$ be a sample from some distribution with probability function f and parameter θ. To test

$$H_0 : \theta = \theta_0 \qquad H_A : \theta = \theta_1,$$

the **Neyman-Pearson** test rejects H_0 when

$$\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} = \frac{f(X_1, ..., X_n|\theta_1)}{f(X_1, ..., X_n|\theta_0)} > k$$

where k is chosen so that $\beta(\theta_0) = \alpha$.

- Wald test relies on asymptotic normality
- A likelihood ratio test can be used for any distribution, provided I can write down the likelihood!